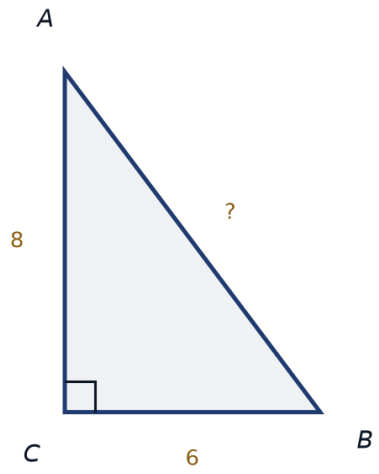
Geometry, Triangles & Right-Triangle Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

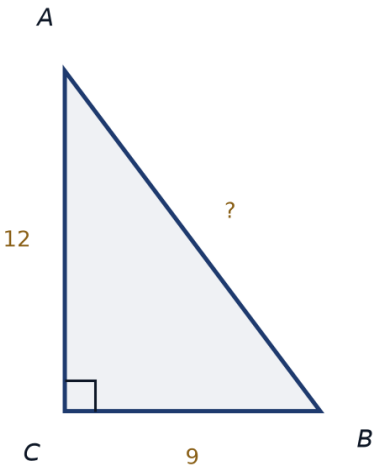
1 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the municipal library, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



2 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

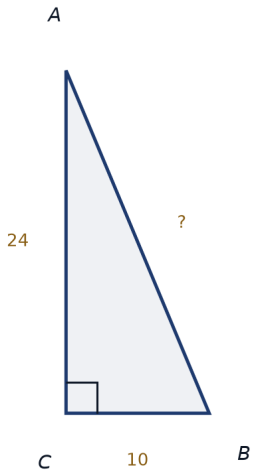
In right triangle ABC shown for a support frame at the coastal aquarium, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



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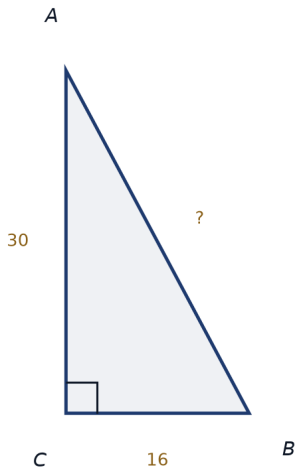
3 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the urban bike-share station, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



4 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

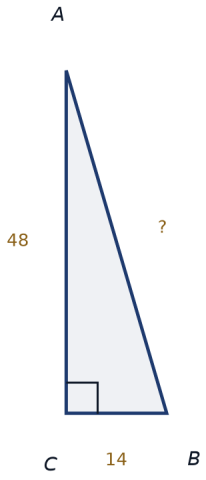
In right triangle ABC shown for a support frame at the agricultural cooperative, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



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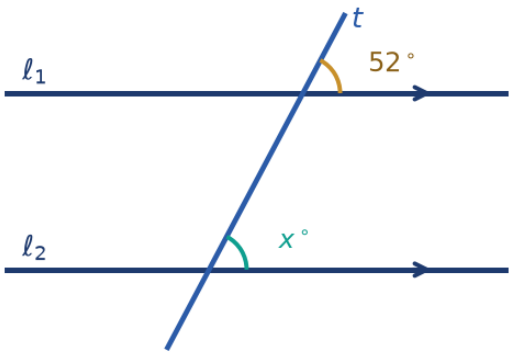
5 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the recycling center, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



6 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

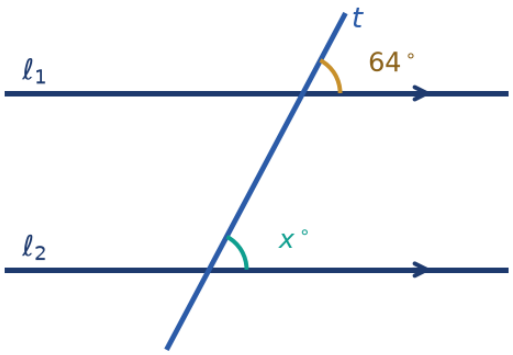
In the figure shown for the roof truss at the neighborhood bookstore, parallel lines ℓ and m are intersected by a transversal line. Based on the angle measure labeled on the diagram, what is the value of x (where x° is the supplementary interior angle)?



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7 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the figure shown for the roof truss at the culinary institute, parallel lines ℓ and m are intersected by a transversal line. Based on the angle measure labeled on the diagram, what is the value of x (where x° is the supplementary interior angle)?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the orchestra hall, in the circle with center O shown for the courtyard at the campus coffee roastery, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

9 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the circle with center O shown for the courtyard at the honeybee apiary, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

10 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the local farmers market, in the circle with center O shown for the courtyard at the historic lighthouse, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

11 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the performing arts academy, in the circle with center O shown for the courtyard at the local farmers market, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

12 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the coastal aquarium, in the circle with center O shown for the courtyard at the neighborhood hardware store, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

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13 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the public transit depot, in a calculation for the community theater (case 2), a circle has equation $(x-2)^2+(y+1)^2=16$. What is the radius?

14 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the wildlife sanctuary, in the coordinate plane, point A is at $(1,2)$ and point B is at $(5,5)$. What is AB?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the historic lighthouse, in a calculation for the charter ferry service (case 38), a right triangle has legs of length 7 and 24. What is the hypotenuse?

16 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the botanical conservatory, $\sin(18^\circ)=\cos(x^\circ)$, where $0<x<90$. What is the value of x?

17 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the municipal library layout, the equation $x^2-6x+y^2+8y=0$ defines a circle. What is the radius of the circle?

18 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A cylindrical water column at the community theater has a base radius of 5 feet and a height of 7 feet. If the volume of the cylinder is $k\pi$ cubic feet, what is the value of k?

19 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the regional bakery, $\sin(21^\circ)=\cos(x^\circ)$, where $0<x<90$. What is the value of x?

20 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

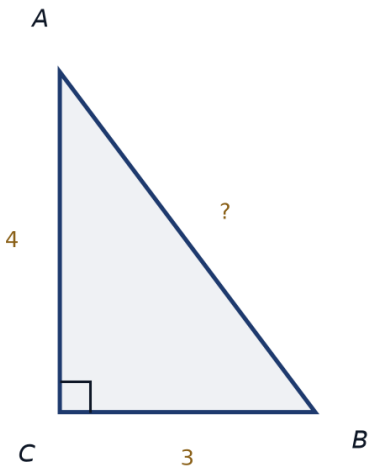
In the xy-plane for the coastal aquarium layout, the equation $x^2-12x+y^2+14y=-21$ defines a circle. What is the radius of the circle?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the botanical conservatory, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



a. 5

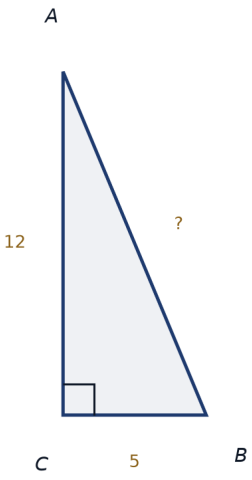
b. 7

c. 4

d. 8

22 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the community theater, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



a. 17

b. 12

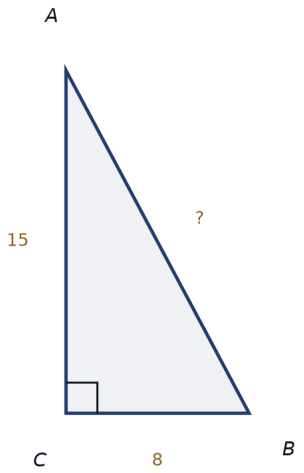
c. 13

d. 15

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23 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the regional bakery, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



a. 59

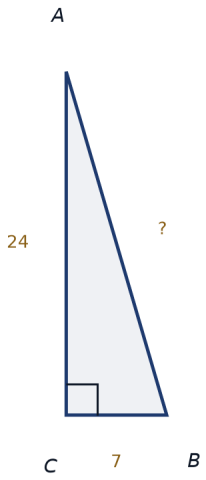
b. 60

c. 62

d. 23

24 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the mountain observatory, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



a. 86

b. 31

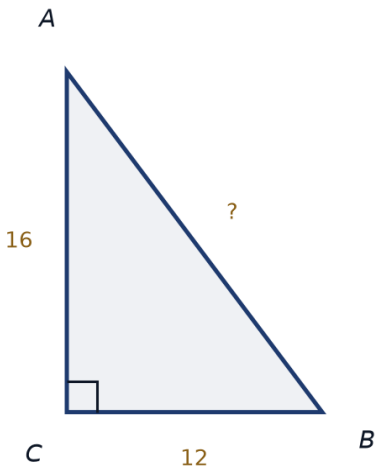
c. 83

d. 84

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25 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the science museum, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



a. 28

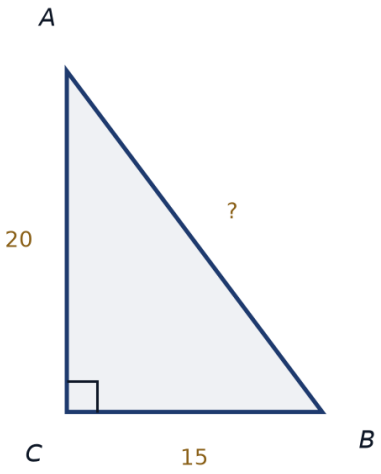
b. 19

c. 20

d. 22

26 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the solar microgrid facility, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?



a. 25

b. 27

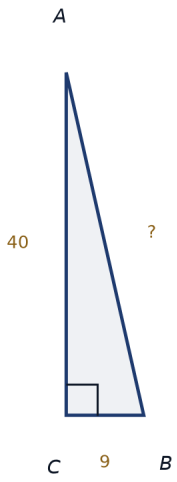
c. 35

d. 24

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27 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the ceramic studio, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



a. 182

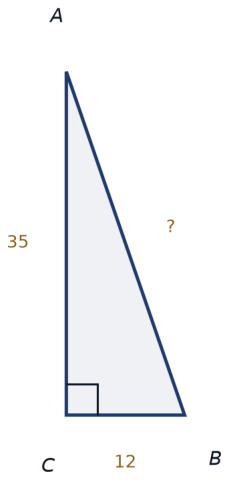
b. 49

c. 179

d. 180

28 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the publishing house, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?



a. 209

b. 210

c. 212

d. 47

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Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig**

In a model for the wildlife sanctuary, in the circle with center O shown for the courtyard at the environmental club, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 324

b. 162

c. 72

d. 360

30 **[Intermediate]** • **[MCQ]** • **Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig**

In a model for the charter ferry service, in the circle with center O shown for the courtyard at the neighborhood hardware store, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 36

b. 27

c. 24

d. 12

31 **[Intermediate]** • **[MCQ]** • **Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig**

In a model for the textile workshop, in the circle with center O shown for the courtyard at the aquatic center, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 168

b. 144

c. 72

d. 48

32 **[Intermediate]** • **[MCQ]** • **Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig**

In a model for the geology lab, in the circle with center O shown for the courtyard at the archery range, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 432

b. 216

c. 72

d. 468

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33 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the carpentry workshop, in the circle with center O shown for the courtyard at the wind energy cooperative, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 48

b. 36

c. 18

d. 24

34 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the historic lighthouse, in the circle with center O shown for the courtyard at the carpentry workshop, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 108

b. 54

c. 36

d. 126

35 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the community orchard, in the circle with center O shown for the courtyard at the marine research vessel, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 48

b. 120

c. 99

d. 96

36 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the environmental club, in the circle with center O shown for the courtyard at the national park visitor center, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 51

b. 48

c. 24

d. 60

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37 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the neighborhood hardware store, in the circle with center O shown for the courtyard at the community orchard, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 54

b. 27

c. 36

d. 72

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the honeybee apiary, in a calculation for the wildlife sanctuary (case 36), what is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. 96 cm²

c. 128 cm²

d. 144 cm²

39 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the carpentry workshop, in a circle, a central angle of 60° intercepts what fraction of the entire circumference?

a. $\frac{1}{4}$

b. $\frac{1}{5}$

c. $\frac{1}{6}$

d. $\frac{1}{8}$

40 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the community orchard, in a calculation for the textile workshop (case 39), what is the area of a parallelogram with base 12 cm and height 5 cm?

a. 30 cm²

b. 60 cm²

c. 90 cm²

d. 120 cm²

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41 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the local farmers market, a rectangle has perimeter 36 cm and length = twice width. What is the area in cm²?

a. 36

b. 48

c. 64

d. 72

42 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the orchestra hall layout, the equation $x^2 - 12x + y^2 + 10y = 39$ defines a circle. What is the radius of the circle?

a. 11

b. 12

c. 10

d. 100

43 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A cylindrical water column at the sports complex has a base radius of 6 feet and a height of 12 feet. If the volume of the cylinder is $k\pi$ cubic feet, what is the value of k?

a. 438

b. 432

c. 144

d. 72

44 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the charter ferry service, $\sin(42^\circ) = \cos(x^\circ)$, where $0 < x < 90$. What is the value of x?

a. 48

b. 42

c. 138

d. 58

45 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the maple syrup farm layout, the equation $x^2 - 6x + y^2 + 6y = 7$ defines a circle. What is the radius of the circle?

a. 25

b. 6

c. 7

d. 5

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